INFOSYS OA + INTERVIEW DSA PREPARATION LIST

1. ARRAY

Basic

- Searching
- Sorting
- Maximum / Minimum element
- Frequency count
- Prefix Sum
- Sliding Window basics

Subarray Problems

- Maximum Subarray Sum (Kadane's Algorithm)
- Subarray with Given Sum
- Count Subarray with Given Sum
- Longest Subarray with Sum K
- Number of Nice Subarrays
- Sum of all Subarrays
- Prefix Sum + HashMap concepts

Binary Search on Answer

- Koko Eating Bananas
- Aggressive Cows
- Search in Rotated Sorted Array
- First and Last Occurrence
- Peak Element
- Allocate Minimum Pages

Two Pointer / Greedy Array

- Move Zeroes to End
- Sort Colors (Dutch National Flag)
- Majority Element
- Merge Intervals
- Non-overlapping Intervals
- Buy & Sell Stock I, II
- Trapping Rain Water
- Container With Most Water
- Merge Two Sorted Arrays
- Next Permutation

---

2. STRING

Basic

- Input / Output
- String Functions
- Reverse String
- Palindrome Check

Hashing + Map Based

- Valid Anagram
- Isomorphic String
- Pangram
- K-Anagram

Sliding Window on String

- Longest Substring Without Repeating Characters
- Longest Substring with K Distinct Characters
- Minimum Window Substring
- Permutation in String

Advanced String

- Longest Common Prefix
- Remove Outermost Parentheses
- Sort Characters by Frequency
- Longest Palindromic Substring
- Sum of Beauty of All Substrings

---

3. RECURSION + BACKTRACKING

Basic Recursion

- Print Subsequences
- Print Subsets
- Subset Sum
- Generate Parentheses

Backtracking

- N Queen
- Sudoku Solver
- Combination Sum I & II
- Word Search
- Rat in a Maze
- Palindrome Partitioning
- Permutations
- Letter Combination of Phone Number

---

4. STACK & QUEUE

Basic

- Stack Implementation
- Queue Implementation
- STL / Predefined Functions

Important Problems

- Valid Parentheses
- Min Stack
- Next Greater Element
- Next Smaller Element
- Stock Span Problem
- Largest Rectangle in Histogram
- Sliding Window Maximum
- Asteroid Collision
- Celebrity Problem
- Implement Queue using Stack
- Implement Stack using Queue

---

5. GREEDY

- Jump Game I & II
- Job Sequencing
- Activity Selection
- Minimum Platforms
- Fractional Knapsack

- Candy Distribution
- Gas Station
- Line Sweep Technique

---

6. LINKED LIST

Basic

- Singly Linked List
- Doubly Linked List
- Circular Linked List

Important Problems

- Reverse Linked List
- Detect Cycle
- Find Middle Node
- Merge Two Sorted Lists
- Remove Nth Node from End
- Palindrome Linked List
- Intersection of Linked List

---

7. BINARY TREE

Traversal

- Inorder
- Preorder

- Postorder
- Level Order

Important Problems

- Height / Depth of Binary Tree
- Balanced Binary Tree
- Diameter of Tree
- Maximum Path Sum
- Lowest Common Ancestor (LCA)
- Print Nodes at Distance K
- Burn Tree
- Zig Zag Traversal
- Vertical Order Traversal
- Boundary Traversal
- Top View / Bottom View
- Left View / Right View

---

8. BST (BINARY SEARCH TREE)

- Search in BST
- Insert Node
- Delete Node
- Kth Smallest Element
- Validate BST
- Lowest Common Ancestor in BST
- Largest BST in Binary Tree

---

9. HEAP / PRIORITY QUEUE

Basic

- Min Heap
- Max Heap
- Heapify

Problems

- Kth Largest Element
- Sort by Frequency
- Merge K Sorted Lists
- Merge K Sorted Arrays
- Top K Frequent Elements
- Minimum Cost to Connect Sticks
- Median in Data Stream

---

10. GRAPH

Basic

- Graph Representation
- BFS
- DFS

Matrix / BFS Problems

- Number of Provinces
- Rotten Oranges
- Flood Fill
- Number of Islands
- Number of Enclaves

Cycle Detection

- Undirected Graph (BFS/DFS)
- Directed Graph

Topological Sort

- Course Schedule
- Alien Dictionary
- Eventual Safe States

Shortest Path

- Dijkstra Algorithm
- Bellman Ford
- Floyd Warshall
- Shortest Path in DAG

Important Problems

- Cheapest Flight Within K Stops
- Number of Ways to Arrive Destination
- City with Smallest Number of Neighbors

---

11. DISJOINT SET UNION (DSU)

- Union by Rank
- Union by Size
- Path Compression

Problems

- Most Stones Removed
- Number of Islands II
- Accounts Merge
- Kruskal Algorithm

---

12. TRIE

- Implement Trie
- Search Suggestions
- Maximum XOR
- Word Dictionary

---

13. DYNAMIC PROGRAMMING (VERY IMPORTANT)

1D DP

- Fibonacci
- Climbing Stairs
- Frog Jump
- House Robber

2D DP

- Unique Paths
- Minimum Path Sum
- Triangle DP

Subsequence DP

- 0/1 Knapsack
- Subset Sum
- Partition Equal Subset Sum
- Coin Change I & II
- Rod Cutting

String DP

- LCS
- Longest Palindromic Subsequence
- Edit Distance
- Distinct Subsequences

LIS Pattern

- Longest Increasing Subsequence
- Largest Divisible Subset
- Bitonic Sequence

Grid + Hard DP

- Cherry Pickup
- Ninja Training
- Burst Balloons
- Matrix Chain Multiplication

---

14. SEGMENT TREE + FENWICK TREE (ADVANCED) - OA

- Range Sum Query
- Point Update
- Lazy Propagation
- Binary Indexed Tree (Fenwick Tree)

---

15. IMPORTANT OA PATTERNS FOR INFOSYS

MOST ASKED PATTERNS

- Sliding Window
- Two Pointer
- Prefix Sum + HashMap
- Binary Search on Answer
- BFS on Grid
- Greedy
- DP on Subsequences
- Heap + Priority Queue
- Backtracking

---

16. INTERVIEW CORE TOPICS

Must Prepare

- Time Complexity
- Space Complexity
- OOPS Concepts
- DBMS Basics
- OS Basics
- CN Basics
- SQL Queries
- Projects Explanation
- Resume Questions

HR Questions

- Tell Me About Yourself
- Strength & Weakness
- Why Infosys?
- Conflict Handling
- Leadership Example
- Internship / Project Discussion

HIGH PRIORITY FOR INFOSYS SP / DSE

MUST DO

- Graph
- DP
- Sliding Window
- Heap
- Binary Search
- Backtracking
- Trees
- Prefix Sum
- HashMap Problems
- String + Window Combination